

Vojtěch Vlček

AI Engineer | LLM Applications & Python

📍 Prague, Czechia

🌐 LinkedIn



🐙 GitHub 📈 Upwork

📄 Summary

AI engineer building LLM applications, AI agents, and Python backend services. Experience owning system architecture and implementation across commercial AI products and university research, including model training, evaluation, and deployment.

</> Technical Skills

Programming languages: Python, Java, C#

AI & machine learning: AI agents, LLM applications, RAG, Model Context Protocol (MCP), LangGraph, Pydantic AI, PyTorch, Hugging Face Diffusers, OpenCV, model training and evaluation.

Development & infrastructure: Linux, Git, Docker, Kubernetes, Microsoft Azure, CUDA, pytest, LangSmith, Langfuse, Weights & Biases.

Backend & data: FastAPI, REST APIs, PostgreSQL, SQLite, pgvector, ETL, data processing.

📁 Professional Experience

Artificial Intelligence Engineer, RESULMATIC

Jul 2026 – Present | Prague, Czechia | Hybrid

- Designed the complete system architecture and implemented most of the backend for an AI automotive assistant covering vehicle search, comparison, and customer enquiries.
- Developed agent workflows and backend services with Python, Pydantic AI, FastAPI, and PostgreSQL, including conversation memory, CRM integration, and handoff to human operators.
- Created an interactive architecture map with React, TypeScript, and MCP integration, enabling developers to explore the system and coding agents to implement selected components in dependency order.
- Automated component verification through tests, type checks, branch coverage, and independent AI code review, with revision workflows and recorded completion evidence.

Applied AI & Deep Learning Engineer, 3BrainAI Nexus

Apr 2026 – Present | Remote | Contract

- Develop AI and deep learning solutions for construction risk assessment, with broad technical responsibility across the project.

Freelance AI Engineer, Upwork Clients 

2025 – Jul 2026 | Remote

- Delivered freelance AI development and generative-media projects, translating client requirements into applications and automated content-generation workflows.
- Developed image and video generation workflows using Python, FastAPI, Gemini, Veo, and Kling, including prompt pipelines, background processing, and automated branding.
- Managed client communication and project delivery across assignments, earning 5/5 client ratings on Upwork.

Primary Researcher, Sign Language Synthesis, University of West Bohemia
University-affiliated research project.

Sep 2024 – Sep 2025 | Pilsen, Czechia

- Designed and built a new sign-language synthesis pipeline combining computer vision and generative AI. Owned the full development process, from system design and data preparation through training, evaluation, and deployment.
- Developed custom pose-conditioned models using PyTorch, Hugging Face Diffusers, and Stable Diffusion.
- Used Kubernetes for model training and deployment, Docker and Singularity for reproducible environments, and Weights & Biases for experiment tracking.

Contract Python Developer, Czech Academy of Sciences

Mar 2025 – Sep 2025 | Prague, Czechia

- Designed and built a Python application used daily by researchers to process experimental datasets and prepare them for analysis.
- Integrated automated batch conversion, data validation, and synchronization of experimental events to produce consistent datasets for legacy research software.

Contract Researcher, *University of West Bohemia*

Sep 2024 – Jan 2025 | Pilsen, Czechia

- Researched and developed computer-vision solutions for automated video analysis using Python and OpenCV.
- Translated experimental methods into reusable software, combining image analysis, parallel processing, and automated quality assessment.

Contract Python Developer, *Czech Academy of Sciences*

Jan 2022 – Jan 2023 | Prague, Czechia

- Built, tested, and debugged a Python memory-test application for a Czech Academy of Sciences research team.

Selected Projects

CleanCodeRefactor  , *AI Code Refactoring*

- Designed and built an AI-assisted refactoring system that analyzes Python codebases, generates improvements, and verifies proposed changes.
- Combined code analysis, retrieval of coding guidance, automated test generation, and AI review into a coordinated development workflow.
- Implemented isolated workspaces, rollback, and resumable execution, with Langfuse tracing and change reports for reviewing results before applying them.

AutoChannel  , *AI Video Production*

- Designed and built an end-to-end AI video-production system, automating scriptwriting, narration, visual generation, video assembly, and YouTube publishing.
- Created a modular Python architecture integrating language, speech, image, and video models, with interchangeable providers including OpenAI, Google, and Kling.
- Implemented resumable production workflows, centralized prompt management, and automated rendering using SQLite, Langfuse, MoviePy, and FFmpeg.

MuAlphaZeroLibrary 

- Built a PyTorch library for AlphaZero and MuZero with self-play training.

JavaTorch 

- Built a Java deep-learning framework prototype with automatic differentiation.

Education

Computer Science, *University of West Bohemia*

Aug 2024 – Aug 2025 | Pilsen, Czechia

Undergraduate studies in Computer Science.

High School Diploma, *Gymnázium Da Vinci*

2020 – 2024 | Czechia

Training

Deep Learning  , *Neuromatch Academy*

Jul 2023

- Completed an intensive three-week deep learning program combining hands-on PyTorch training with a collaborative research project.

Machine Learning Specialization  , *Stanford Online & DeepLearning.AI, via Coursera*

Oct 2022 – Nov 2022

- Completed a three-course specialization covering supervised and unsupervised learning, neural networks, recommender systems, and reinforcement learning.

Languages

English: Proficient (C2) • **Czech:** Native